

Nishan Shetty MS, MBA

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EDUCATION

Master of Business Administration , Population Health Management – Ottawa University	May 2024
Master of Science , Industrial Engineering – Arizona State University	May 2019

CERTIFICATIONS

NVIDIA-Certified Associate, Generative AI LLMs | **AWS**, Certified Machine Learning – Specialty | **Databricks**, Machine Learning Engineer – Associate | **AZ-900**, Microsoft Azure Fundamentals

SUMMARY

Senior Data Scientist & AI Engineer with 8+ years designing and deploying production ML systems across GenAI, agentic AI, NLP, and classical machine learning in regulated environments. Deep expertise in RAG pipelines, multi-agent orchestration, clinical NLP, time-series forecasting, and causal experimentation, with a strong track record translating complex business problems into scalable AI solutions on AWS and Azure. Equally comfortable owning the full technical stack and mentoring junior engineers.

SKILLS

Programming: Python | SQL | R

Agentic AI & LLMs: LangChain | LangGraph | Multi-agent orchestration | Tool orchestration | Function calling | Prompt engineering | RAG pipeline design | Vector stores (FAISS, Elastic, DynamoDB) | Hybrid search | Re-ranking | OpenAI APIs | AWS Bedrock | Azure AI Studio | HuggingFace | GPT | Claude | Llama

NLP & Classical ML: spaCy | scispaCy | BERT | word2vec | Flair | Transformers | Entity extraction | Text classification | Semantic similarity | Negation handling | Topic modeling | Embeddings | Classification | Regression | Clustering | Time Series (ARIMA, SARIMA, Prophet, LSTM) | Uplift modeling | Causal inference | A/B testing | Anomaly detection | Recommender systems

MLOps & LLMops: MLflow | SageMaker | Azure ML | Docker | Kubernetes | CI/CD | Airflow | Lambda | Step Functions | Drift detection | Retraining pipelines | Versioning and rollback

Data Engineering: Spark | Databricks | Snowflake | ETL pipelines | Kafka | PyTorch | TensorFlow | Keras | FastAPI | Git

WORK EXPERIENCE

Senior Data Scientist Southwest Medical Imaging Arizona	December 2018 – Present
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Agentic AI & GenAI Systems

- Designed and deployed multi-agent AI systems using LangChain and LangGraph for autonomous task execution, routing outputs to downstream workflows and flagging exceptions for human review, cutting manual review effort by 45% and increasing follow-up compliance by 30%.
- Built end-to-end RAG pipelines with chunking, embedding, and vector retrieval layers using FAISS, Elastic, and DynamoDB via SageMaker and Bedrock, implementing hybrid search and re-ranking to achieve grounding and precision levels needed for compliance-critical production deployment.
- Built tool-using agents with function calling and secure API integrations connecting to scheduling, compliance, and reporting systems, enabling self-service decision support and reducing manual reporting burden by more than 50%.

LLM Evaluation, Prompt Engineering & LLMops

- Defined evaluation frameworks covering task success, factuality, grounding, latency, and cost, building observability into agent behavior and tool call failure modes so issues were caught before reaching end users or auditors.
- Applied prompt engineering, instruction tuning, and LoRA fine-tuning to adapt foundation models for domain-specific use cases, using feedback-driven optimization loops to improve instruction-following and output reliability in production.

NLP & Clinical Text Analytics

- Built clinical NLP pipelines using spaCy, scispaCy, BERT, and HuggingFace Transformers for entity extraction, negation handling, section detection, and text classification on radiology and clinical documentation corpora, producing structured outputs from high-volume unstructured text.

- Applied semantic similarity, topic modeling, and embedding-based retrieval to surface relevant clinical signals from large document sets, enabling downstream compliance and reporting workflows to operate on structured data rather than raw text.

Classical ML, Forecasting & Experimentation

- Built time-series forecasting models (ARIMA, LSTM, hybrid ensembles) on SageMaker predicting claim denial risk and A/R aging weeks in advance, improving forecast accuracy by 18%, reducing denials by 25%, and cutting reprocessing costs by 77%.
- Built segmentation models using K-Means, hierarchical clustering, and PCA with causal uplift modeling and A/B experimentation to measure true intervention impact, improving campaign ROI by 25%.

MLOps & LLMOps

- Built CI/CD-driven MLOps pipelines with Docker, MLflow, Git, SageMaker, and Airflow, enabling automated drift detection, retraining, versioning, and rollback across production ML systems and LLM applications deployed across 25+ sites.
- Implemented automated retraining pipelines with Lambda-triggered nightly jobs, S3-based feature storage, and production monitoring that kept forecasting and classification models accurate over time without manual intervention.
- Engineered scalable data and ML pipelines using Python, SQL, Databricks, Spark, Snowflake, and Airflow, supporting ingestion, transformation, model deployment, and recurring analytics delivery across all sites.

Principal Data Scientist

September 2018 – September 2019

eTrack Tech | Remote

- Built predictive maintenance models across a 10,000+ device IoT fleet that identified failure risk before it materialized, reducing fault-driven downtime by 22%.
- Used NLP to extract and classify root cause patterns from unstructured technician logs, automating a previously manual process and integrating outputs directly into operational workflows.

Graduate Research Assistant

January 2018 – August 2018

Mayo Clinic – Arizona State University Alliance | Arizona

- Performed exploratory analyses and predictive model evaluation in collaboration with clinical researchers, adhering to IRB-aligned protocols and healthcare data compliance standards.
- Championed AI literacy through documentation, internal knowledge-sharing sessions, and collaboration with academic partners.

Product Manager

August 2016 – August 2017

Danfoss | Dubai

- Built financial and risk models for energy solutions, defining product vision and technical roadmaps that improved energy efficiency outcomes for each client.
- Partnered with engineering and analytics teams to build A/B testing frameworks and deliver measurable adoption gains across customer segments.

KEY PROJECTS

Incidental Findings Follow-up Agent (Multi-tool Agentic RAG)

- Manual tracking of findings across high report volumes created compliance risk and patient safety gaps. Built a LangChain and Bedrock RAG agent that read reports, assessed urgency, and autonomously triggered follow-up and compliance workflows via secure APIs, with DynamoDB retrieval and full audit logging throughout.
- Increased follow-up compliance by 30%, reduced manual review effort by 45%.

Client & Provider Segmentation with Agentic Experimentation

- Outreach campaigns were generic and unmeasured, with no way to distinguish what was actually driving behavior change. Built a framework combining K-Means clustering, causal uplift modeling, and recommender systems that segmented by behavioral profile, ran controlled experiments, and automatically personalized outreach per segment.
- Improved campaign ROI by 25%, giving leadership causal evidence for operational decisions rather than correlation.

Accounts Receivable & Claim Denial Forecasting

- Finance teams were reacting to revenue gaps rather than anticipating them. Built ARIMA, LSTM, and hybrid ensemble forecasting models on SageMaker with Lambda-triggered nightly retraining, integrating predictions into Snowflake dashboards for proactive scenario planning.
- Reduced reprocessing costs by 77%, lowered claim denials by 25%, improved forecast accuracy by 18%.